MASTER SYSTEM DESIGN TRACKER - 380 CONCEPTS

Master tracker for all 380 indexed concepts. Learned = core idea is correct. +1 and +7 = recalled after one and seven days. Explain = stated the decision, trade-off, and one failure without notes.

ID	Question / concept	Learned	+1	+7	Explain
01 Distributed Systems Fundamentals					
1	Distributed systems	[ ]	[ ]	[ ]	[ ]
2	CAP theorem	[ ]	[ ]	[ ]	[ ]
3	PACELC theorem	[ ]	[ ]	[ ]	[ ]
4	Consistency models	[ ]	[ ]	[ ]	[ ]
5	Strong consistency	[ ]	[ ]	[ ]	[ ]
6	Eventual consistency	[ ]	[ ]	[ ]	[ ]
7	Causal consistency	[ ]	[ ]	[ ]	[ ]
8	Read-your-writes consistency	[ ]	[ ]	[ ]	[ ]
9	Linearizability	[ ]	[ ]	[ ]	[ ]
10	Serializability	[ ]	[ ]	[ ]	[ ]
11	Consensus algorithms	[ ]	[ ]	[ ]	[ ]
12	Raft consensus	[ ]	[ ]	[ ]	[ ]
13	Paxos	[ ]	[ ]	[ ]	[ ]
14	Leader election	[ ]	[ ]	[ ]	[ ]
15	Split-brain problem	[ ]	[ ]	[ ]	[ ]
16	Quorum reads and writes	[ ]	[ ]	[ ]	[ ]
17	Distributed locks	[ ]	[ ]	[ ]	[ ]
18	Distributed transactions	[ ]	[ ]	[ ]	[ ]
19	Two-phase commit (2PC)	[ ]	[ ]	[ ]	[ ]
20	Saga pattern	[ ]	[ ]	[ ]	[ ]
21	Idempotency keys	[ ]	[ ]	[ ]	[ ]
22	Exactly-once vs at-least-once vs at-most-once processing	[ ]	[ ]	[ ]	[ ]
23	Logical clocks	[ ]	[ ]	[ ]	[ ]
24	Vector clocks	[ ]	[ ]	[ ]	[ ]
25	Lamport timestamps	[ ]	[ ]	[ ]	[ ]
02 Scalability & Load Balancing					
26	Horizontal scaling	[ ]	[ ]	[ ]	[ ]
27	Vertical scaling	[ ]	[ ]	[ ]	[ ]
28	Stateless vs stateful services	[ ]	[ ]	[ ]	[ ]
29	Load balancing	[ ]	[ ]	[ ]	[ ]
30	Layer 4 vs Layer 7 load balancers	[ ]	[ ]	[ ]	[ ]
31	Round-robin load balancing	[ ]	[ ]	[ ]	[ ]
32	Least-connections routing	[ ]	[ ]	[ ]	[ ]
33	Consistent hashing	[ ]	[ ]	[ ]	[ ]
34	Sticky sessions	[ ]	[ ]	[ ]	[ ]
35	Reverse proxies	[ ]	[ ]	[ ]	[ ]
36	Forward proxies	[ ]	[ ]	[ ]	[ ]
37	API gateways	[ ]	[ ]	[ ]	[ ]
38	Service discovery	[ ]	[ ]	[ ]	[ ]
39	Health checks	[ ]	[ ]	[ ]	[ ]
40	Connection pooling	[ ]	[ ]	[ ]	[ ]
41	Rate limiting	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
42	Token bucket algorithm	[ ]	[ ]	[ ]	[ ]
43	Leaky bucket algorithm	[ ]	[ ]	[ ]	[ ]
44	Fixed-window rate limiting	[ ]	[ ]	[ ]	[ ]
45	Sliding-window rate limiting	[ ]	[ ]	[ ]	[ ]
46	Backpressure	[ ]	[ ]	[ ]	[ ]
47	Load shedding	[ ]	[ ]	[ ]	[ ]
48	Autoscaling	[ ]	[ ]	[ ]	[ ]
49	Hotspots and hot partitions	[ ]	[ ]	[ ]	[ ]
50	Thundering herd problem	[ ]	[ ]	[ ]	[ ]
03 Caching & Content Delivery					
51	Cache-aside pattern	[ ]	[ ]	[ ]	[ ]
52	Read-through caching	[ ]	[ ]	[ ]	[ ]
53	Write-through caching	[ ]	[ ]	[ ]	[ ]
54	Write-behind caching	[ ]	[ ]	[ ]	[ ]
55	Cache invalidation	[ ]	[ ]	[ ]	[ ]
56	Cache eviction policies (LRU, LFU, FIFO)	[ ]	[ ]	[ ]	[ ]
57	Cache TTL	[ ]	[ ]	[ ]	[ ]
58	Cache stampede	[ ]	[ ]	[ ]	[ ]
59	Cache penetration	[ ]	[ ]	[ ]	[ ]
60	Cache avalanche	[ ]	[ ]	[ ]	[ ]
61	Edge caching	[ ]	[ ]	[ ]	[ ]
62	Content Delivery Networks (CDNs)	[ ]	[ ]	[ ]	[ ]
63	Browser caching	[ ]	[ ]	[ ]	[ ]
64	HTTP caching headers	[ ]	[ ]	[ ]	[ ]
65	ETag and conditional requests	[ ]	[ ]	[ ]	[ ]
66	Redis caching	[ ]	[ ]	[ ]	[ ]
67	Distributed caching	[ ]	[ ]	[ ]	[ ]
68	Cache consistency	[ ]	[ ]	[ ]	[ ]
69	Negative caching	[ ]	[ ]	[ ]	[ ]
70	Cache warming	[ ]	[ ]	[ ]	[ ]
04 Databases & Data Modeling					
71	SQL vs NoSQL	[ ]	[ ]	[ ]	[ ]
72	ACID properties	[ ]	[ ]	[ ]	[ ]
73	BASE properties	[ ]	[ ]	[ ]	[ ]
74	Database normalization	[ ]	[ ]	[ ]	[ ]
75	Database denormalization	[ ]	[ ]	[ ]	[ ]
76	Primary keys vs foreign keys	[ ]	[ ]	[ ]	[ ]
77	Indexes	[ ]	[ ]	[ ]	[ ]
78	Composite indexes	[ ]	[ ]	[ ]	[ ]
79	Covering indexes	[ ]	[ ]	[ ]	[ ]
80	B-tree vs B+ tree	[ ]	[ ]	[ ]	[ ]
81	Hash indexes	[ ]	[ ]	[ ]	[ ]
82	Query execution plans	[ ]	[ ]	[ ]	[ ]
83	N+1 query problem	[ ]	[ ]	[ ]	[ ]
84	Database sharding	[ ]	[ ]	[ ]	[ ]
85	Database partitioning	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
86	Database replication	[ ]	[ ]	[ ]	[ ]
87	Leader-follower replication	[ ]	[ ]	[ ]	[ ]
88	Multi-leader replication	[ ]	[ ]	[ ]	[ ]
89	Read replicas	[ ]	[ ]	[ ]	[ ]
90	Replication lag	[ ]	[ ]	[ ]	[ ]
91	Optimistic locking	[ ]	[ ]	[ ]	[ ]
92	Pessimistic locking	[ ]	[ ]	[ ]	[ ]
93	MVCC (Multi-Version Concurrency Control)	[ ]	[ ]	[ ]	[ ]
94	Transaction isolation levels	[ ]	[ ]	[ ]	[ ]
95	Deadlocks	[ ]	[ ]	[ ]	[ ]
96	Database connection pools	[ ]	[ ]	[ ]	[ ]
97	Materialized views	[ ]	[ ]	[ ]	[ ]
98	Change Data Capture (CDC)	[ ]	[ ]	[ ]	[ ]
99	CQRS	[ ]	[ ]	[ ]	[ ]
100	Event sourcing	[ ]	[ ]	[ ]	[ ]
05 Messaging, Queues & Event-Driven Architecture					
101	Message queues	[ ]	[ ]	[ ]	[ ]
102	Publish-subscribe pattern	[ ]	[ ]	[ ]	[ ]
103	Kafka topics and partitions	[ ]	[ ]	[ ]	[ ]
104	Consumer groups	[ ]	[ ]	[ ]	[ ]
105	Consumer offsets	[ ]	[ ]	[ ]	[ ]
106	Message acknowledgments	[ ]	[ ]	[ ]	[ ]
107	Dead-letter queues (DLQs)	[ ]	[ ]	[ ]	[ ]
108	Retry queues	[ ]	[ ]	[ ]	[ ]
109	Exponential backoff	[ ]	[ ]	[ ]	[ ]
110	Jitter	[ ]	[ ]	[ ]	[ ]
111	Poison messages	[ ]	[ ]	[ ]	[ ]
112	Message deduplication	[ ]	[ ]	[ ]	[ ]
113	Ordering guarantees	[ ]	[ ]	[ ]	[ ]
114	Outbox pattern	[ ]	[ ]	[ ]	[ ]
115	Inbox pattern	[ ]	[ ]	[ ]	[ ]
116	Event-driven architecture	[ ]	[ ]	[ ]	[ ]
117	Eventual consistency in messaging	[ ]	[ ]	[ ]	[ ]
118	Event replay	[ ]	[ ]	[ ]	[ ]
119	Stream processing	[ ]	[ ]	[ ]	[ ]
120	Queue visibility timeout	[ ]	[ ]	[ ]	[ ]
06 Networking, DNS & HTTP					
121	OSI model	[ ]	[ ]	[ ]	[ ]
122	TCP/IP model	[ ]	[ ]	[ ]	[ ]
123	TCP three-way handshake	[ ]	[ ]	[ ]	[ ]
124	TCP vs UDP	[ ]	[ ]	[ ]	[ ]
125	DNS resolution	[ ]	[ ]	[ ]	[ ]
126	DNS caching	[ ]	[ ]	[ ]	[ ]
127	DNS TTL	[ ]	[ ]	[ ]	[ ]
128	DNS load balancing	[ ]	[ ]	[ ]	[ ]
129	HTTP/1.1 vs HTTP/2 vs HTTP/3	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
130	HTTPS and TLS	[ ]	[ ]	[ ]	[ ]
131	TLS handshake	[ ]	[ ]	[ ]	[ ]
132	SSL termination / TLS termination	[ ]	[ ]	[ ]	[ ]
133	HTTP keep-alive	[ ]	[ ]	[ ]	[ ]
134	Connection reuse	[ ]	[ ]	[ ]	[ ]
135	WebSockets	[ ]	[ ]	[ ]	[ ]
136	Server-Sent Events (SSE)	[ ]	[ ]	[ ]	[ ]
137	Long polling	[ ]	[ ]	[ ]	[ ]
138	gRPC	[ ]	[ ]	[ ]	[ ]
139	REST APIs	[ ]	[ ]	[ ]	[ ]
140	GraphQL	[ ]	[ ]	[ ]	[ ]
141	HTTP status codes	[ ]	[ ]	[ ]	[ ]
142	HTTP request methods	[ ]	[ ]	[ ]	[ ]
143	CORS	[ ]	[ ]	[ ]	[ ]
144	NAT	[ ]	[ ]	[ ]	[ ]
145	Anycast routing	[ ]	[ ]	[ ]	[ ]
07 Microservices & API Design					
146	Monolith vs microservices	[ ]	[ ]	[ ]	[ ]
147	Modular monolith	[ ]	[ ]	[ ]	[ ]
148	Service-oriented architecture (SOA)	[ ]	[ ]	[ ]	[ ]
149	API versioning	[ ]	[ ]	[ ]	[ ]
150	API pagination	[ ]	[ ]	[ ]	[ ]
151	Cursor-based pagination	[ ]	[ ]	[ ]	[ ]
152	Offset-based pagination	[ ]	[ ]	[ ]	[ ]
153	API idempotency	[ ]	[ ]	[ ]	[ ]
154	API request validation	[ ]	[ ]	[ ]	[ ]
155	API aggregation	[ ]	[ ]	[ ]	[ ]
156	Backend for Frontend (BFF)	[ ]	[ ]	[ ]	[ ]
157	Circuit breaker pattern	[ ]	[ ]	[ ]	[ ]
158	Bulkhead pattern	[ ]	[ ]	[ ]	[ ]
159	Timeout management	[ ]	[ ]	[ ]	[ ]
160	Retry policies	[ ]	[ ]	[ ]	[ ]
161	Service mesh	[ ]	[ ]	[ ]	[ ]
162	Sidecar pattern	[ ]	[ ]	[ ]	[ ]
163	Distributed tracing	[ ]	[ ]	[ ]	[ ]
164	Correlation IDs	[ ]	[ ]	[ ]	[ ]
165	Contract testing	[ ]	[ ]	[ ]	[ ]
166	Consumer-driven contracts	[ ]	[ ]	[ ]	[ ]
167	API schema evolution	[ ]	[ ]	[ ]	[ ]
168	Feature flags	[ ]	[ ]	[ ]	[ ]
169	Strangler fig pattern	[ ]	[ ]	[ ]	[ ]
170	Anti-corruption layer	[ ]	[ ]	[ ]	[ ]
08 Kubernetes, Containers & Deployment					
171	Containers vs virtual machines	[ ]	[ ]	[ ]	[ ]
172	Docker images	[ ]	[ ]	[ ]	[ ]
173	Multi-stage Docker builds	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
174	Container registries	[ ]	[ ]	[ ]	[ ]
175	Kubernetes Pods	[ ]	[ ]	[ ]	[ ]
176	Kubernetes Deployments	[ ]	[ ]	[ ]	[ ]
177	Kubernetes ReplicaSets	[ ]	[ ]	[ ]	[ ]
178	Kubernetes StatefulSets	[ ]	[ ]	[ ]	[ ]
179	Kubernetes DaemonSets	[ ]	[ ]	[ ]	[ ]
180	Kubernetes Services	[ ]	[ ]	[ ]	[ ]
181	ClusterIP, NodePort, LoadBalancer	[ ]	[ ]	[ ]	[ ]
182	Kubernetes Ingress	[ ]	[ ]	[ ]	[ ]
183	Gateway API	[ ]	[ ]	[ ]	[ ]
184	ConfigMaps	[ ]	[ ]	[ ]	[ ]
185	Kubernetes Secrets	[ ]	[ ]	[ ]	[ ]
186	Liveness probes	[ ]	[ ]	[ ]	[ ]
187	Readiness probes	[ ]	[ ]	[ ]	[ ]
188	Startup probes	[ ]	[ ]	[ ]	[ ]
189	Horizontal Pod Autoscaler (HPA)	[ ]	[ ]	[ ]	[ ]
190	Vertical Pod Autoscaler (VPA)	[ ]	[ ]	[ ]	[ ]
191	Kubernetes resource requests and limits	[ ]	[ ]	[ ]	[ ]
192	PersistentVolumes (PV)	[ ]	[ ]	[ ]	[ ]
193	PersistentVolumeClaims (PVC)	[ ]	[ ]	[ ]	[ ]
194	Helm charts	[ ]	[ ]	[ ]	[ ]
195	Helm values and templates	[ ]	[ ]	[ ]	[ ]
196	Rolling deployments	[ ]	[ ]	[ ]	[ ]
197	Blue-green deployments	[ ]	[ ]	[ ]	[ ]
198	Canary deployments	[ ]	[ ]	[ ]	[ ]
199	Kubernetes Jobs and CronJobs	[ ]	[ ]	[ ]	[ ]
200	PodDisruptionBudgets (PDB)	[ ]	[ ]	[ ]	[ ]
09 Reliability, Resilience & Fault Tolerance					
201	Fault tolerance	[ ]	[ ]	[ ]	[ ]
202	High availability	[ ]	[ ]	[ ]	[ ]
203	Redundancy	[ ]	[ ]	[ ]	[ ]
204	Single point of failure (SPOF)	[ ]	[ ]	[ ]	[ ]
205	Graceful degradation	[ ]	[ ]	[ ]	[ ]
206	Fail-fast principle	[ ]	[ ]	[ ]	[ ]
207	Failover and failback	[ ]	[ ]	[ ]	[ ]
208	Active-active architecture	[ ]	[ ]	[ ]	[ ]
209	Active-passive architecture	[ ]	[ ]	[ ]	[ ]
210	Disaster recovery	[ ]	[ ]	[ ]	[ ]
211	Recovery Time Objective (RTO)	[ ]	[ ]	[ ]	[ ]
212	Recovery Point Objective (RPO)	[ ]	[ ]	[ ]	[ ]
213	Multi-region architecture	[ ]	[ ]	[ ]	[ ]
214	Multi-availability-zone architecture	[ ]	[ ]	[ ]	[ ]
215	Chaos engineering	[ ]	[ ]	[ ]	[ ]
216	Fault injection	[ ]	[ ]	[ ]	[ ]
217	Retry storms	[ ]	[ ]	[ ]	[ ]
218	Cascading failures	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
219	Heartbeats	[ ]	[ ]	[ ]	[ ]
220	Failure detection	[ ]	[ ]	[ ]	[ ]
221	Timeout propagation	[ ]	[ ]	[ ]	[ ]
222	Request hedging	[ ]	[ ]	[ ]	[ ]
223	Retry budgets	[ ]	[ ]	[ ]	[ ]
224	Graceful shutdown	[ ]	[ ]	[ ]	[ ]
225	Zero-downtime deployments	[ ]	[ ]	[ ]	[ ]
10 Observability & Production Monitoring					
226	Logging vs metrics vs tracing	[ ]	[ ]	[ ]	[ ]
227	Structured logging	[ ]	[ ]	[ ]	[ ]
228	Centralized logging	[ ]	[ ]	[ ]	[ ]
229	Distributed tracing	[ ]	[ ]	[ ]	[ ]
230	Trace IDs and span IDs	[ ]	[ ]	[ ]	[ ]
231	OpenTelemetry	[ ]	[ ]	[ ]	[ ]
232	Prometheus	[ ]	[ ]	[ ]	[ ]
233	Grafana	[ ]	[ ]	[ ]	[ ]
234	Splunk	[ ]	[ ]	[ ]	[ ]
235	CloudWatch	[ ]	[ ]	[ ]	[ ]
236	Application Performance Monitoring (APM)	[ ]	[ ]	[ ]	[ ]
237	Service Level Indicators (SLIs)	[ ]	[ ]	[ ]	[ ]
238	Service Level Objectives (SLOs)	[ ]	[ ]	[ ]	[ ]
239	Service Level Agreements (SLAs)	[ ]	[ ]	[ ]	[ ]
240	Error budgets	[ ]	[ ]	[ ]	[ ]
241	Golden signals (latency, traffic, errors, saturation)	[ ]	[ ]	[ ]	[ ]
242	RED method (Rate, Errors, Duration)	[ ]	[ ]	[ ]	[ ]
243	USE method (Utilization, Saturation, Errors)	[ ]	[ ]	[ ]	[ ]
244	Alert fatigue	[ ]	[ ]	[ ]	[ ]
245	Alert thresholds	[ ]	[ ]	[ ]	[ ]
246	Percentile latency (P50, P95, P99)	[ ]	[ ]	[ ]	[ ]
247	Incident management	[ ]	[ ]	[ ]	[ ]
248	Root cause analysis (RCA)	[ ]	[ ]	[ ]	[ ]
249	Postmortems	[ ]	[ ]	[ ]	[ ]
250	Synthetic monitoring	[ ]	[ ]	[ ]	[ ]
11 Security, Authentication & Authorization					
251	Authentication vs authorization	[ ]	[ ]	[ ]	[ ]
252	OAuth 2.0	[ ]	[ ]	[ ]	[ ]
253	OpenID Connect (OIDC)	[ ]	[ ]	[ ]	[ ]
254	JWT	[ ]	[ ]	[ ]	[ ]
255	Session-based authentication	[ ]	[ ]	[ ]	[ ]
256	API keys	[ ]	[ ]	[ ]	[ ]
257	Role-Based Access Control (RBAC)	[ ]	[ ]	[ ]	[ ]
258	Attribute-Based Access Control (ABAC)	[ ]	[ ]	[ ]	[ ]
259	Least privilege principle	[ ]	[ ]	[ ]	[ ]
260	Zero Trust architecture	[ ]	[ ]	[ ]	[ ]
261	Mutual TLS (mTLS)	[ ]	[ ]	[ ]	[ ]
262	Secrets management	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
263	Key rotation	[ ]	[ ]	[ ]	[ ]
264	Encryption at rest	[ ]	[ ]	[ ]	[ ]
265	Encryption in transit	[ ]	[ ]	[ ]	[ ]
266	Symmetric vs asymmetric encryption	[ ]	[ ]	[ ]	[ ]
267	Hashing vs encryption	[ ]	[ ]	[ ]	[ ]
268	Salting and password hashing	[ ]	[ ]	[ ]	[ ]
269	SQL injection	[ ]	[ ]	[ ]	[ ]
270	Cross-Site Scripting (XSS)	[ ]	[ ]	[ ]	[ ]
271	Cross-Site Request Forgery (CSRF)	[ ]	[ ]	[ ]	[ ]
272	Server-Side Request Forgery (SSRF)	[ ]	[ ]	[ ]	[ ]
273	Distributed Denial of Service (DDoS)	[ ]	[ ]	[ ]	[ ]
274	Web Application Firewalls (WAF)	[ ]	[ ]	[ ]	[ ]
275	Certificate management	[ ]	[ ]	[ ]	[ ]
12 Cloud Infrastructure & DevOps					
276	Infrastructure as Code (IaC)	[ ]	[ ]	[ ]	[ ]
277	Terraform	[ ]	[ ]	[ ]	[ ]
278	AWS CloudFormation	[ ]	[ ]	[ ]	[ ]
279	CI/CD pipelines	[ ]	[ ]	[ ]	[ ]
280	GitHub Actions	[ ]	[ ]	[ ]	[ ]
281	Jenkins	[ ]	[ ]	[ ]	[ ]
282	GitOps	[ ]	[ ]	[ ]	[ ]
283	Argo CD	[ ]	[ ]	[ ]	[ ]
284	Immutable infrastructure	[ ]	[ ]	[ ]	[ ]
285	Infrastructure drift	[ ]	[ ]	[ ]	[ ]
286	Environment promotion	[ ]	[ ]	[ ]	[ ]
287	Artifact versioning	[ ]	[ ]	[ ]	[ ]
288	Deployment rollback	[ ]	[ ]	[ ]	[ ]
289	AWS IAM	[ ]	[ ]	[ ]	[ ]
290	AWS VPC	[ ]	[ ]	[ ]	[ ]
291	Public vs private subnets	[ ]	[ ]	[ ]	[ ]
292	Security groups	[ ]	[ ]	[ ]	[ ]
293	Route tables	[ ]	[ ]	[ ]	[ ]
294	Internet gateways vs NAT gateways	[ ]	[ ]	[ ]	[ ]
295	AWS Lambda	[ ]	[ ]	[ ]	[ ]
296	AWS S3	[ ]	[ ]	[ ]	[ ]
297	AWS RDS	[ ]	[ ]	[ ]	[ ]
298	AWS SQS and SNS	[ ]	[ ]	[ ]	[ ]
299	AWS EventBridge	[ ]	[ ]	[ ]	[ ]
300	AWS Step Functions	[ ]	[ ]	[ ]	[ ]
13 Concurrency & Multithreading					
301	Processes vs threads	[ ]	[ ]	[ ]	[ ]
302	Concurrency vs parallelism	[ ]	[ ]	[ ]	[ ]
303	Race conditions	[ ]	[ ]	[ ]	[ ]
304	Critical sections	[ ]	[ ]	[ ]	[ ]
305	Mutex locks	[ ]	[ ]	[ ]	[ ]
306	Semaphores	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
307	Read-write locks	[ ]	[ ]	[ ]	[ ]
308	Reentrant locks	[ ]	[ ]	[ ]	[ ]
309	Thread pools	[ ]	[ ]	[ ]	[ ]
310	Thread starvation	[ ]	[ ]	[ ]	[ ]
311	Deadlock prevention	[ ]	[ ]	[ ]	[ ]
312	Livelock	[ ]	[ ]	[ ]	[ ]
313	Atomic operations	[ ]	[ ]	[ ]	[ ]
314	Compare-and-swap (CAS)	[ ]	[ ]	[ ]	[ ]
315	Lock-free programming	[ ]	[ ]	[ ]	[ ]
316	Memory visibility	[ ]	[ ]	[ ]	[ ]
317	Happens-before relationship	[ ]	[ ]	[ ]	[ ]
318	Context switching	[ ]	[ ]	[ ]	[ ]
319	Producer-consumer problem	[ ]	[ ]	[ ]	[ ]
320	Thread-safe data structures	[ ]	[ ]	[ ]	[ ]
14 Storage, Files & Data Processing					
321	Object vs block vs file storage	[ ]	[ ]	[ ]	[ ]
322	Blob storage	[ ]	[ ]	[ ]	[ ]
323	Object versioning	[ ]	[ ]	[ ]	[ ]
324	Multipart uploads	[ ]	[ ]	[ ]	[ ]
325	Presigned URLs	[ ]	[ ]	[ ]	[ ]
326	Data compression	[ ]	[ ]	[ ]	[ ]
327	Data deduplication	[ ]	[ ]	[ ]	[ ]
328	Checksums	[ ]	[ ]	[ ]	[ ]
329	Data integrity	[ ]	[ ]	[ ]	[ ]
330	Write-Ahead Logging (WAL)	[ ]	[ ]	[ ]	[ ]
331	Log-Structured Merge Trees (LSM Trees)	[ ]	[ ]	[ ]	[ ]
332	Bloom filters	[ ]	[ ]	[ ]	[ ]
333	Merkle trees	[ ]	[ ]	[ ]	[ ]
334	Consistent snapshots	[ ]	[ ]	[ ]	[ ]
335	Incremental backups	[ ]	[ ]	[ ]	[ ]
336	Batch processing	[ ]	[ ]	[ ]	[ ]
337	Stream processing	[ ]	[ ]	[ ]	[ ]
338	ETL vs ELT	[ ]	[ ]	[ ]	[ ]
339	Data lakes	[ ]	[ ]	[ ]	[ ]
340	Data warehouses	[ ]	[ ]	[ ]	[ ]
15 Advanced Architecture & Performance					
341	Domain-Driven Design (DDD)	[ ]	[ ]	[ ]	[ ]
342	Bounded contexts	[ ]	[ ]	[ ]	[ ]
343	Hexagonal architecture	[ ]	[ ]	[ ]	[ ]
344	Clean architecture	[ ]	[ ]	[ ]	[ ]
345	Dependency inversion	[ ]	[ ]	[ ]	[ ]
346	Twelve-Factor App	[ ]	[ ]	[ ]	[ ]
347	Command Query Separation (CQS)	[ ]	[ ]	[ ]	[ ]
348	Fan-out / fan-in	[ ]	[ ]	[ ]	[ ]
349	Scatter-gather pattern	[ ]	[ ]	[ ]	[ ]
350	MapReduce	[ ]	[ ]	[ ]	[ ]

ID	Question / concept	Learned	+1	+7	Explain
351	Work stealing	[ ]	[ ]	[ ]	[ ]
352	Async vs sync communication	[ ]	[ ]	[ ]	[ ]
353	Request coalescing	[ ]	[ ]	[ ]	[ ]
354	Batching	[ ]	[ ]	[ ]	[ ]
355	Micro-batching	[ ]	[ ]	[ ]	[ ]
356	Little's Law	[ ]	[ ]	[ ]	[ ]
357	Queueing theory	[ ]	[ ]	[ ]	[ ]
358	Amdahl's Law	[ ]	[ ]	[ ]	[ ]
359	Latency vs throughput	[ ]	[ ]	[ ]	[ ]
360	Capacity planning	[ ]	[ ]	[ ]	[ ]
16 Testing, Release Engineering & Compatibility					
361	Unit testing	[ ]	[ ]	[ ]	[ ]
362	Integration testing	[ ]	[ ]	[ ]	[ ]
363	End-to-end testing	[ ]	[ ]	[ ]	[ ]
364	Load testing	[ ]	[ ]	[ ]	[ ]
365	Stress testing	[ ]	[ ]	[ ]	[ ]
366	Soak testing	[ ]	[ ]	[ ]	[ ]
367	Spike testing	[ ]	[ ]	[ ]	[ ]
368	Contract testing	[ ]	[ ]	[ ]	[ ]
369	Chaos testing	[ ]	[ ]	[ ]	[ ]
370	Shadow traffic	[ ]	[ ]	[ ]	[ ]
371	Traffic mirroring	[ ]	[ ]	[ ]	[ ]
372	Dark launches	[ ]	[ ]	[ ]	[ ]
373	A/B testing	[ ]	[ ]	[ ]	[ ]
374	Backward compatibility	[ ]	[ ]	[ ]	[ ]
375	Forward compatibility	[ ]	[ ]	[ ]	[ ]
376	Expand-and-contract database migrations	[ ]	[ ]	[ ]	[ ]
377	Schema migration strategies	[ ]	[ ]	[ ]	[ ]
378	Rolling upgrades	[ ]	[ ]	[ ]	[ ]
379	Deployment health gates	[ ]	[ ]	[ ]	[ ]
380	Automated rollback	[ ]	[ ]	[ ]	[ ]

MORNING REVIEW LOG

Record the missed idea, date, concept IDs reviewed, and how many of five answers were correct before reading.

Missed idea / repair	Date	Concept IDs	Score / 5